Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory were checked in Lean, where the formal version differs from the brief, and any problems found in the brief. The brief is available at `https://assumptionsofphysics.org/resources/briefs`.

Setup

- Lean `v4.36.0-rc2`; Mathlib at commit `1b1ccf4b`; physlib (`master`) and TauCeti (commit `e7e7f52`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Equivalence of sup-norm, L^1, and L^2 seminorms on the Schwartz space

Files.

- File: `briefs/SchwartzNorms/SchwartzNorms.lean`
- Library: `SchwartzNorms`
- Imports: `SchwartzMomentProblem` (for the moments in eq. (10)) and three new `ForMathlib` files:
  `ForMathlib/Analysis/LocallyConvex/WithSeminorms.lean` (bounds between seminorm families compose),
  `ForMathlib/Analysis/SpecialFunctions/ImproperIntegrals.lean` ($\int (1+x^2)^{-2} = \pi/2$, $\int_{|x|>1} x^{-2} = 2$),
  `ForMathlib/MeasureTheory/Integral/IntegralEqImproper.lean` ($\|g(x)\| \le \|g'\|_{L^1}$ for integrable g, g')

Setting. For $f : \mathbb{R} \to \mathbb{C}$ and $g = x^\alpha \partial^\beta f$ (`weighted`), the seminorms (1)–(3) are `supNorm`, `l1Norm` and `l2Norm`, with values in $[0, \infty]$ (`eLpNorm` for L^1 and L^2). Smoothness is `ContDiff` of order ∞. On $\mathcal{S}(\mathbb{R})$, the L^1 and L^2 seminorms are real seminorms `l1Seminorm`, `l2Seminorm`, collected in the families `l1Family`, `l2Family`; the sup family is Mathlib's `schwartzSeminormFamily`.

Results.

Brief	Lean
eq. (6), sup controls L^2	`l2Norm_sq_le`
eq. (7), L^2 controls L^1	`l1Norm_le`
eq. (9), L^1 controls sup	`supNorm_le` (with $\|f\|^{L^1}_{\alpha,\beta} < \infty$); `not_supNorm_le_const` (the hypothesis is needed)
Theorem 5, (i) $\Leftrightarrow$ (ii) $\Leftrightarrow$ (iii)	`tfae_finite`
definition (4) is Mathlib's $\mathcal{S}(\mathbb{R})$	`exists_schwartzMap_iff`, `supNorm_eq_seminorm`
Step 4, same topology	`withSeminorms_l1_l2`, from `isBounded_sup_l2`, `isBounded_l2_l1`, `isBounded_l1_sup` and `WithSeminorms.congr_of_cycle`
eq. (10)	`enorm_moment_le` (with the factor $\|\hbar\|^\beta$)

Differences from the brief.

- The seminorms take values in $[0, \infty]$, so that Theorem 5 can be stated for every smooth f. On $\mathcal{S}(\mathbb{R})$ they are finite and give real seminorms.
- "Generate the same locally convex topology" is stated as `WithSeminorms`: each family induces Mathlib's topology on $\mathcal{S}(\mathbb{R})$. Step 4's "this relation is transitive" is `Seminorm.IsBounded.comp`, new in `ForMathlib`.
- In Step 3, $g(x) = \int_{-\infty}^{x} g'$ uses Mathlib's improper FTC and the fact that an integrable g with integrable g' tends to 0 at $-\infty$.
- The converse in the remark after eq. (10) (finite expectation values imply membership in $\mathcal{S}(\mathbb{R})$) and the remark on general L^p are not formalized.

Problems found in the brief.

- Eq. (9) needs $\|f\|^{L^1}_{\alpha,\beta} < \infty$, so that $g \to 0$ at $-\infty$. The brief has it from (ii), but (9) is stated as an inequality between seminorms, and it is false without it: for $f = 1$, $\alpha = \beta = 0$, the left side is 1 and the right side is 0.
- In eq. (7) the coefficient of $\|f\|^{L^2}_{\alpha+1,\beta}$ is 2; the argument given ($\sqrt{a+b+c} \le \sqrt{a} + \sqrt{b} + \sqrt{c}$ with $b = 2(\|f\|^{L^2}_{\alpha+1,\beta})^2$) gives $\sqrt{2}$. The stated bound is still true, and is the one formalized.
- Eq. (10) omits the factor $|\hbar|^\beta$ from $P^\beta = (-i\hbar\partial)^\beta$.